



A restatement of the case for speed limits

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ABSTRACT

This paper provides a restatement of the case for speed limits. The paper argues that driver speed choice cannot be granted any normative status (i.e. be regarded as optimal from a societal point of view) unless it is “objectively” rational, even if it can be reasonably interpreted as “subjectively” rational. A distinction between “subjective” and “objective” rationality is very rarely made in modern analyses relying on the theory of rational choice, but it makes sense with respect to the choice of speed. Studies that have assessed which impacts of speed drivers consider, whether drivers correctly assess these impacts, and how drivers coordinate their choice of speed with other drivers are reviewed. It is concluded that driver speed choice is not “objectively” rational. It is concluded that the lack of rationality in driver speed choice implies that this choice needs to be regulated by means of speed limits.

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1. Introduction

Today, speed limits are almost universal. Although there are still sections on German Autobahns that do not have speed limits, speed limits are spreading even on these last remnants of a road system with no speed limits. It is taken for granted that there should be speed limits. It would therefore seem to be an anachronism to raise the question of why there should be speed limits and what the principal arguments for them are. This paper gives a restatement of the case for speed limits. The case is based on a critical examination of driver rationality in speed choice. The argument is made that speed limits are needed to efficiently coordinate driver speed choice based on normative criteria of rationality, as there are reasons to believe that a “free” choice of speed would not produce outcomes that are optimal from a societal point of view.

The choice of speed can be studied from several different perspectives. For the purpose of the analysis presented in this paper, the perspective of normative economic welfare theory has been chosen. One reason for this choice is that normative welfare economics is based on a respect for citizen sovereignty (often referred to as consumer sovereignty in economic theory) and a principle of methodological individualism (i.e. the point of view that all social phenomena originate in individual behaviour and choices) (Adler and Posner, 2001, 2006; Mishan and Quah, 2007). In other words, the analysis is based on actual individual

behaviour, e.g. the actual choices of speed that road users make. A second reason for adopting an economic perspective is that this perspective represents a clear normative ideal for individual behaviour, in terms of social efficiency. Normative economic welfare theory proposes social efficiency as an ideal solution to any problem that involves interaction between individuals. A solution is regarded as socially efficient if it is impossible to improve the welfare of one person without reducing the welfare of another person. Such an outcome is often referred to as Pareto-optimal.

As far as speed is concerned, a Pareto-optimal solution will be defined as the level of speed which is optimal from a societal perspective (Elvik, 2002). Optimal speed is the speed that minimises the total costs of travel. Any speed that deviates from the optimal level will reduce welfare by increasing costs, irrespective of who pays these additional costs. The main question to be asked is therefore whether a free choice of speed is likely to result in speeds that are optimal from a societal perspective or not. If the answer to this question is negative, regulation of speed choice is necessary, subject to the remediableness criterion proposed by Williamson (2000), in order to bring actual choices closer to the optimal level. More specifically, speed limits will be seen as needed if:

1. Driver speed choice gives rise to external effects, i.e. effects that influence the well-being of others, but are not taken into consideration by drivers.
2. Drivers base their choices of speed on an incorrect perception of the impacts of speed, e.g. an underestimation of the effects on accident risk.

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3. Drivers have heterogeneous preferences with respect to speed, making the coordination of speed choice between drivers difficult.

These conditions, in particular the presence of externalities, have traditionally been treated as “market imperfections” in economic welfare theory and been used to justify regulations that constrain individual behaviour or the introduction of taxes that influence individual incentives.

It should be noted that the use and determination of speed limits can be analysed from different perspectives than the one adopted in this paper. From a public health perspective, which underlies the Vision Zero ideal for transport safety (Vägverket, 1997), speed limits should be set so that impact speed in accidents never exceeds human tolerance for biomechanical impacts that can lead to death or permanent impairment. In other words, impacts resulting in death or permanent impairment should be eliminated and speed limits should comply with this basic design parameter for the transport system. It will be noted that road user preferences, as well as other impacts of speed, are completely ignored in this approach.

Another perspective is the traffic engineering perspective. According to this perspective, speed limits should reflect the design standard of a road: a freeway, designed to the highest standards, should have a high-speed limit, whereas an access road serving only local traffic in a residential area should have a low-speed limit. One version of this perspective is the so-called 85th percentile criterion, which has traditionally been used in the United States. According to this criterion, speed limits should be set at the speed that 85 percent of drivers comply with. This essentially amounts to accepting the speeds drivers choose as the guiding criterion for setting speed limits.

The choice of an economic perspective in this paper is in no way intended to suggest that other perspectives are less informative. It does, however, permit an analysis of speed choice in terms of clearly stated criteria of rationality. The application of the theory of rational choice to the analysis of driver speed choice is discussed below.

2. Application of the theory of rational choice to driver speed choice

Modern theory of rational choice defines rationality in terms of the beliefs and preferences of the individual. It is, as Jon Elster states (2007, p. 209) “subjective through and through”, i.e. it refers only to what individuals believe and prefer and not to some external standard. Elster adds that: “One might, to be sure, take the word “rational” in an objective sense, implying that a rational agent is one who makes decisions that make his life go better as judged by objective criteria such as health, longevity, or income. Used in this way, however, the idea would not have any explanatory power.” This point of view is obviously correct as far as explaining choices by showing that they were (subjectively) rational is concerned.

Herbert Simon (1976, p. 76), on the other hand, makes the following distinction between subjective and objective rationality: “A decision may be called “objectively” rational if in fact it is the correct behaviour for maximizing given values in a given situation. It is “subjectively” rational if it maximizes attainment relative to the actual knowledge of the subject.”

The distinction made by Simon is used as the starting point for the analysis presented in this paper. Throughout the paper, the choice of speed is treated as subjectively rational. Speed choice and the perceptions underlying it will be modelled formally in terms of continuous functions. An obvious objection to such

models is that they do not reflect the way speed choice is perceived and experienced by each driver. This criticism is correct, but any formal model should be interpreted in “as if” terms, i.e. driver behaviour can be modelled as if it obeyed the functions proposed here. A model which is much closer to what drivers mentally experience has been proposed by Vaa (2003, 2007). His model is intended as an explanatory model, and has a different analytical purpose from the models employed here.

3. Assessing the rationality of driver speed choice

To assess the rationality of driver speed choice, relevant research is reviewed in this section. This research consists of three types of studies:

1. Studies of the factors that influence speed choice and of the importance drivers assign to these factors.
2. Studies of how drivers perceive the relationship between speed and travel time, speed and accident risk and speed and stopping distance.
3. Studies of the speed at which drivers prefer to travel and models of the coordination of speed choice between drivers.

Driver speed choice will not be regarded as objectively rational to the extent it is not based on a consideration of all impacts of speed, is based on an erroneous perception of these impacts, or leads to a traffic speed that many drivers would not freely have chosen.

3.1. Factors influencing speed

Several studies have examined factors that influence speed choice (Nilsson, 1991; Rajalin and Summala, 1996; Åberg et al., 1997; Haglund and Åberg, 2000; Letirand and Delhomme, 2005; Stradling, 2007; Wallén Warner and Åberg, 2008; Schmid Mast et al., 2008; Tarko, 2009). Of particular interest are studies that identified multiple impacts of speed and tried to determine the importance of these impacts for driver speed choice.

Nilsson (1991) asked a sample of drivers why they were not driving faster. A total of 12 reasons were given and drivers rated the importance of these reasons. Among drivers who complied with the speed limit, acceptance of the speed limit was the most important reason for not driving faster. Increase in emissions was rated at place eight (out of 12) and increased noise was rated at place 11. Among drivers who exceeded the speed limit by less than 10 km/h, increase of emissions was rated at place eight and increased noise at place 11 out of the 12 reasons stated. Among drivers who exceeded the speed limit by more than 10 km/h, increase of emissions and noise were the two least important reasons they gave for not driving faster. In other words, consideration of impacts on noise and emissions does not prevent a driver from driving fast and are, even among those who comply with speed limits, rated as the least important factors influencing speed choice. Nilsson also asked drivers why they were not driving slower. The most important reason given was that driving slower would impede traffic. Among the fastest drivers, dullness was also given as an important reason for not driving slower.

Rajalin and Summala (1996) obtained slightly different results. They asked a sample of slow drivers why they were driving slowly. Fifteen reasons were listed. The most important reason given for driving slowly was that the driver had plenty of time. Consideration for car, fuel and environment was rated at sixth place among the 15 reasons, suggesting that it does carry some

weight among slow drivers. It was, however, clearly not the most important reason for driving slowly.

Stradling (2007) found that large majorities of drivers stated that they would drive slower in fog, in heavy rain, in darkness or on unfamiliar roads. Interestingly, 69% of drivers also stated that they would drive slower if traffic was slower than their normal speed. 55% stated that they would speed up if late for a meeting or an appointment; 30% stated that they would speed up if traffic was faster than their normal speed. Impacts of speed choice on noise or pollution were not considered in this study.

Wallén Warner and Åberg (2008) found that drivers regarded it as likely that exceeding speed limits in urban areas would increase pollution and rated this impact as very bad. This knowledge and evaluation did not, however, correlate very strongly with driver intention to exceed speed limits. Among 10 factors, it was rated at place seven in terms of the strength of the correlation with the intention to exceed speed limits in urban areas. With respect to the intention to exceed speed limits in rural areas, it was once more found that drivers believed that this would increase pollution and regarded this impact as very undesirable. This belief and evaluation had only minor influence on their intention to speed. The correlation between the assessment of impacts on pollution and intention to speed was only a non-significant .07, the eleventh weakest correlation of 14 factors that were included.

Taken together, these studies suggest that while drivers know that speed influences noise and pollution, and regard an increase in noise and pollution as undesirable, consideration of these impacts carry little weight in determining the choice of speed. It is reasonable to assume that impacts of speed choice on noise and pollution are to a large extent external from the driver's point of view.

Neglecting impacts on noise and pollution may have significant implications for speed choice. In an analysis of optimal speed limits, Elvik (2002) found that impacts on noise and pollution represent about 16 percent of the total societal costs of travel at a speed of 50 km/h in urban areas and about 4 percent of the total societal costs of travel at a speed of 80 km/h in rural areas. Official Norwegian monetary valuations of travel time, accidents, noise and air pollution were applied (Eriksen et al., 1999). In urban areas, the costs of noise and air pollution reach their minimum at a speed of 40 km/h.

3.2. Driver perception of impacts of speed on travel time, accident risk and stopping distance

The relationship between mean speed and travel time for a given distance is very simple. It is given by

$$\text{Travel time} = \frac{\text{Distance in km}}{\text{Mean speed in km per hour}}$$

One would believe that the simplicity of this relationship would make it easy to estimate the change in speed required to save a certain amount of time. Svenson (2008, 2009) conducted a series of experiments that show that people do not assess correctly the change in speed required to save a certain amount of travel time. Svenson gave a sample of students at the Royal Institute of Technology in Stockholm a series of tasks that were all framed as follows: "If you are to drive 100 km, increasing your speed from 30 to 40 km/h will save you 50 min of travel time. Suppose your speed is 60 km/h. What speed would you then have to adopt in order to save 50 min?" The correct answer is 120 km/h. If you drive 100 km at 60 km/h, the trip will take 1 h and 40 min; at 120 km/h it will take 50 min, resulting in a saving of 50 min. Surprisingly, students were not able to correctly estimate

Table 1

Subjectively stated increase in speed needed to save the same amount of time as in the reference situation. Based on Svenson (2009).

Reference situation (A)		Situation where new speed is to be stated (B)		
Initial speed (km/h)	New speed (km/h)	Initial speed (km/h)	Stated new speed (km/h)	Correct new speed (km/h)
30	40	60	73	120
40	50	80	95	130
60	120	40	75	60
60	120	30	66	40
30	50	60	88	300
60	130	30	77	40

the changes in speed required to save a certain amount of travel time. Table 1 gives a sample of the results.

In general, when the speed stated in the reference situation was low, and students were asked to estimate the required increase from a high baseline speed, they underestimated how much speed needed to be increased. Conversely, when the speed stated in the reference situation was high, and students were asked to estimate the required increase from a low baseline speed, they overestimated how much speed needed to be increased. In other words, the saving in travel time by small increases of a high speed is overestimated, while the saving in travel time by a small increase in a low speed is underestimated. The relationship between speed and travel time is thus perceived to be more linear than it actually is. Similar results were reproduced in a study by Fuller et al. (2009).

To permit a formal analysis of the implications of the misperception of the relationship between speed and travel time, the results of the studies made by Svenson have been represented in terms of a continuous function. Fig. 1 shows this function as well as the correct relationship between speed and travel time. A trip of 100 km is used as an illustration. The actual relationship between speed and travel time is given by $100/v$ (v =speed in km/h). The perceived relationship is given by

$$\text{Perceived relationship between speed and travel time} = 12.9 \times \exp^{(-0.0255v)} \quad (1)$$

The perceived relationship has a flatter slope than the true relationship for speeds below about 30 km/h. For higher speeds, the slope of the perceived relationship is steeper than for the true relationship. The function for the perceived relationship between speed and travel time is consistent with the findings of Svenson and Fuller et al., but is a smoothing of their findings. The two functions have been calibrated to take on identical values at speeds of 10 and 100 km/h.

It might be objected that the results presented by Svenson and Fuller et al. are hypothetical and apply to the laboratory only. Neither study shows that actual speed choice is influenced by the misperceptions that were found. However, to the extent that the misperceptions found actually influence speed choice, one would expect the value of the ratio: the mean speed of violations/the speed limit to be higher for low-speed limits than for high-speed limits. This is implied by the fact that drivers erroneously think that a major increase in speed is needed to save a certain amount of travel time at low speeds, while a minor increase in speed is sufficient at high speeds. Sørensen et al. (2007) present data for Sweden that support this conjecture. These data are presented in Table 2.

There is a clear tendency for violations of the speed limit, stated as a proportion of the speed limit, to be higher at low speed limits than at high speed limits. This is what one would expect if actual speed choice was influenced by an erroneous perception of

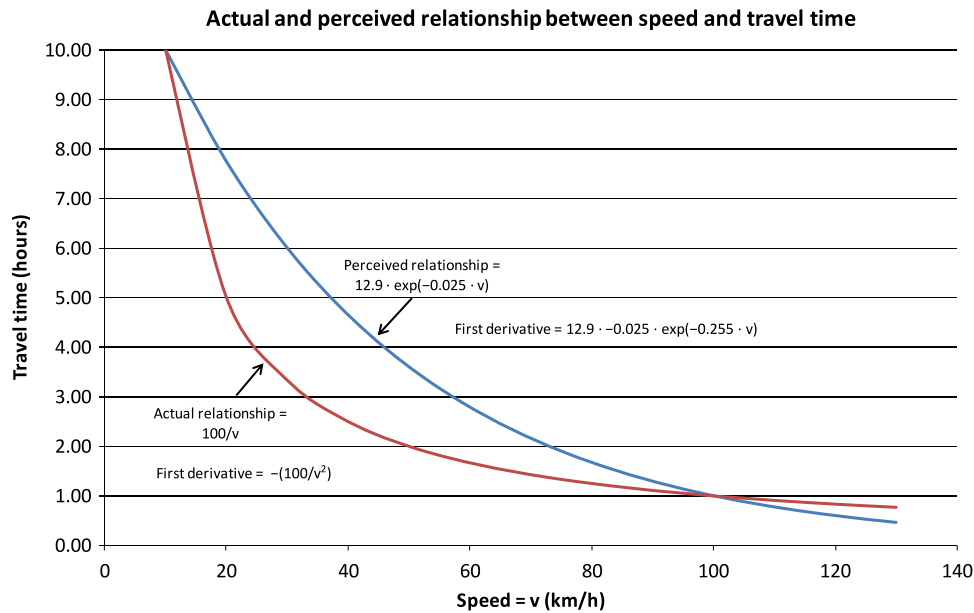


Fig. 1. Actual and perceived relationship between speed and travel time for a given travel distance.

Table 2

Relative level of speed limit violations according to speed limit. Data for Sweden. Based on Sørensen et al. (2007).

Speed limit (km/h)	Mean violation (km/h)	Ratio violation/speed limit
30	10	0.33
50	8	0.16
70	10	0.14
90	11	0.12
110	12	0.11

the relationship between speed and travel time as shown by Svenson and Fuller et al. This may worsen the external effects of speed choice, since low speed limits tend to be found in residential areas where consideration for the safety and comfort of the residents have lead to their introduction. According to the British Crime Survey (Poulter and McKenna, 2007) speeding traffic was rated as the most serious problem of 16 social problems, all of which were rated on a scale from 1 (not a problem at all) to 4 (very big problem).

Svenson (2008, 2009) also asked students to state the increase in the risk of an injury accident associated with specific increases in speed. According to the Power Model of the relationship between speed and road accidents (Elvik et al., 2004), the risk of an injury accident increases in proportion to the second power of the relative change in speed:

$$\frac{\text{Number of accidents after}}{\text{Number of accidents before}} = \left(\frac{\text{Mean speed of traffic after}}{\text{Mean speed of traffic before}} \right)^2$$

A comprehensive meta-analysis of 98 studies containing a total of 460 estimates of the relationship between changes in speed and changes in the number of accidents supported this model (Elvik et al., 2004). Students, however, think that the increase in the risk of an injury accident is much smaller. Fig. 2 shows the relationship between increases in speed and increases in the risk of an injury accident according to the students and according to the Power Model.

It was even more remarkable that students did not think that the risk of a fatal accident increased more than the risk of an injury accident. The Power Model, which is firmly supported by

empirical research, suggests that the risk of a fatal accident increases by about the fourth power of the relative change in speed. Thus, students grossly underestimate both the risk of an accident and accident severity. Svenson asked a sample of traffic engineering students the same questions, but these students were not more well informed than the sample of psychology students that Svenson mainly relied on.

Speed influences both the risk of becoming involved in an accident and the severity of injuries. Both effects can be captured in a single function by means of the Power Model by using exponents that apply to the number of fatalities, the number of seriously injured road users and the number of slightly injured road users. The actual relationship between the changes in speed and the number of killed or injured road users was estimated by means of the following function, using a speed of 100 km/h as reference:

$$\text{Actual effects of changes in speed} = \left(\frac{v_i}{100} \right)^{4.5} + \left(\frac{v_i}{100} \right)^{3.0} + \left(\frac{v_i}{100} \right)^{1.5} \quad (2)$$

100 is the reference speed (km/h); v_i is any other speed (varying between 10 and 130 km/h); 4.5 is the exponent for fatalities; 3.0 is the exponent for seriously injured road users; 1.5 is the exponent for slightly injured road users. Eq. (2) gives relative changes in the number of fatalities, seriously injured road users and slightly injured road users as a result of relative changes in speed. To summarise these changes in terms of a single function, the relative changes were converted to accident costs by applying official monetary valuations for Norway (Statens vegvesen, 2006) and assuming that, as an average for all public roads, 2.25% of injuries are fatal, 8.40% are serious and 89.35% are slight (based on official accident statistics for Norway 2000–2008). The official valuations as of 2005 were 26.5 million NOK for a fatality (1 NOK=0.11 Euro as of July 2009), 7.8 million NOK for a serious injury and 0.8 million NOK for a slight injury. The resulting function, calibrated to equal 1.00 at a speed of 100 km/h was

$$\text{Actual accident costs as a function of speed} = 0.000001 \times v^3 \quad (3)$$

The perceived relationship between speed and the expected costs of accidents was estimated as

$$\text{Perceived accident costs as a function of speed} = 0.002 \times v^{1.21} \quad (4)$$

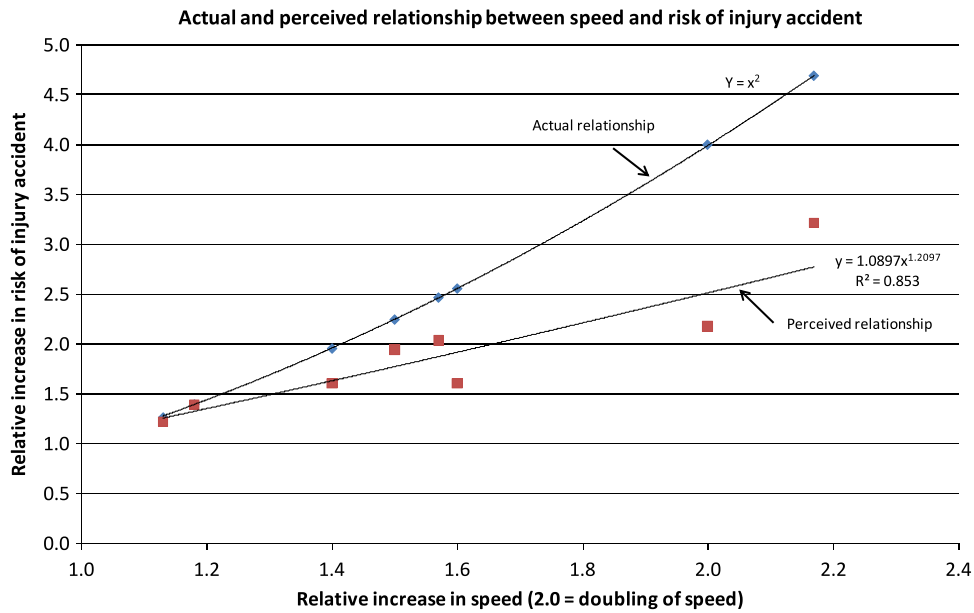


Fig. 2. Increases in the risk of an injury accident according to the Power Model and subjective assessments by students. Based on Svenson (2009).

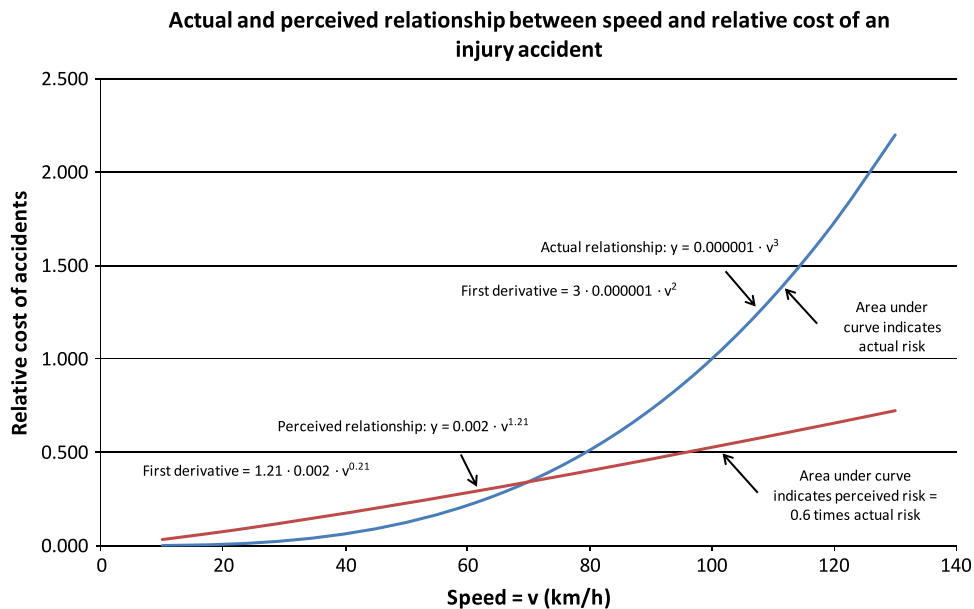


Fig. 3. Actual and perceived relationship between speed and relative accident costs.

The constant term was set so that the area under the curve representing the perceived relationship between speed and relative accident cost was equal to 0.6 times the area under the curve representing the actual relationship between speed and relative accident costs. This is based on a study by Andersson (2007), showing that the median perceived risk of being killed in a road accident is about 60% of the actual risk of being killed (i.e. people underestimate the risk). Fig. 3 shows the two functions.

The third type of task Svenson (2008, 2009) gave students was as follows: “A driver who drives at 25 km/h will just be able to stop before hitting an obstacle that suddenly appears on the road. Your speed is 40 km/h. At what speed do you think you will strike the obstacle?” The most likely actual impact speed was estimated by assuming a reaction time of 1 s (which is short for a surprising situation; Shinar, 2007) and a retardation of 0.8 g. Actual impact

speed was compared to the impact speed estimated by students. Table 3 gives a sample of results.

As can be seen, students underestimated impact speed considerably. If a similar underestimate is prevalent in actual driving, drivers may adopt too small safety margins, erroneously believing that they can stop in a much shorter distance than is actually the case. As an example, driving speeds in darkness are, in general, too high to enable drivers to stop or perform evasive manoeuvres to avoid an accident when only dipped headlights are used and an obstacle suddenly appears (Shinar, 2007).

3.3. The coordination of speed choices

The speed of traffic is the result of the coordination among drivers of their individual speed choices. This coordination

would be easy if all drivers prefer to drive at the same speed. Evidence suggests, however, that this is not the case. There is considerable variation between drivers with respect to preferred speed.

Golbenbeld and van Schagen (2007) assessed the credibility of speed limits by asking a sample of drivers to state: (1) what they regarded as an appropriate speed limit for a sample of 27 road sections, and (2) what they regarded as a safe speed when driving on each of the road sections. All road sections had a speed limit of 80 km/h, but drivers were not informed about this. Safe speed, as stated by drivers, varied between 71 and 92 km/h for the 27 road sections. The study shows that different groups of drivers differ with respect to their views regarding safe speed. In order to explore how preferences regarding safe speed were distributed in the sample of drivers, Goldenbeld and Van Schagen were asked to provide the data for the study, which they kindly did. Fig. 4 is based on driver answers to the question regarding safe speed.

As can be seen, there are two distinct groups among drivers. One large group regarded 80 km/h as a safe speed. Another large group regarded 100 km/h as a safe speed. Preferences regarding safe speed are clearly bimodal, suggesting that the coordination between the two main groups of driver will be difficult: if traffic speed is 80 km/h, many drivers will feel that this is too slow;

if traffic speed is 100 km/h, many drivers will feel that this is too fast.

Based on the distribution of preferences shown in Fig. 4, speed choice can be modelled as a coordination game between two groups of drivers: the fast movers and the slow movers. Table 4 shows this game. The ordinal preferences of drivers are indicated by numbers; 4 is the most preferred outcome, 1 is the least preferred outcome. The fast movers choose between columns; their payoffs are shown in the upper right corner of each cell of the table. The slow movers choose between rows; their payoffs are shown in the bottom left corner of each cell of the table. Both groups of drivers are assumed to prefer driving at the same speed as the other group to driving at a speed that differs from the other group.

For the fast movers, the best outcome is that everybody drives at a 100 km/h. The slow movers will prefer this outcome to driving at 80 km/h, given that the fast movers continue to drive at 100 km/h. However, the most preferred outcome for the slow movers is that everybody drives at 80 km/h. Both this outcome, and the outcome in which everybody drives at 100 km/h, are Nash equilibria, in the sense that no group can get a higher payoff by unilaterally changing its choice. The two solutions are also Pareto-optimal, since going from one solution to the other will reduce the payoff for one group while increasing it for the other group. From a societal point of view, however, the choice between the two solutions is not indifferent. The low-speed equilibrium will be associated with less accidents and smaller environmental impacts than the high-speed equilibrium.

The equilibria are not likely to be stable. Connolly and Åberg (1993) explore a contagion model of speeding, in which speed choice is influenced by other drivers. Depending on how sensitive drivers are to the choices made by other drivers, the mean speed of traffic may be determined by the choices made by even a small minority of drivers. Suppose, for example, that a slow driver maintains a speed of 80 km/h. A fast driver may regard this as too slow, but overtaking is a hassle and it may, on balance, be more pleasant to put up with the slow driver and stay in line. If everybody thinks like that, the speed of traffic will be 80 km/h. A driver who is less sensitive to the speeds chosen by other drivers may decide to overtake the slow driver and may thereby induce

Table 3

Subjectively estimated impact speed for situations in which stopping is impossible. Based on Svenson (2009).

Initial speed for cars that stops (km/h)	Your speed (km/h)	Subjectively stated impact speed (km/h)	Actual impact speed (km/h)
25	40	20.17	40.00
80	90	38.60	44.46
60	130	71.77	130.00
50	70	33.43	57.89
80	120	59.33	99.86
25	50	21.33	50.00
110	130	58.10	74.38
70	110	57.83	95.96

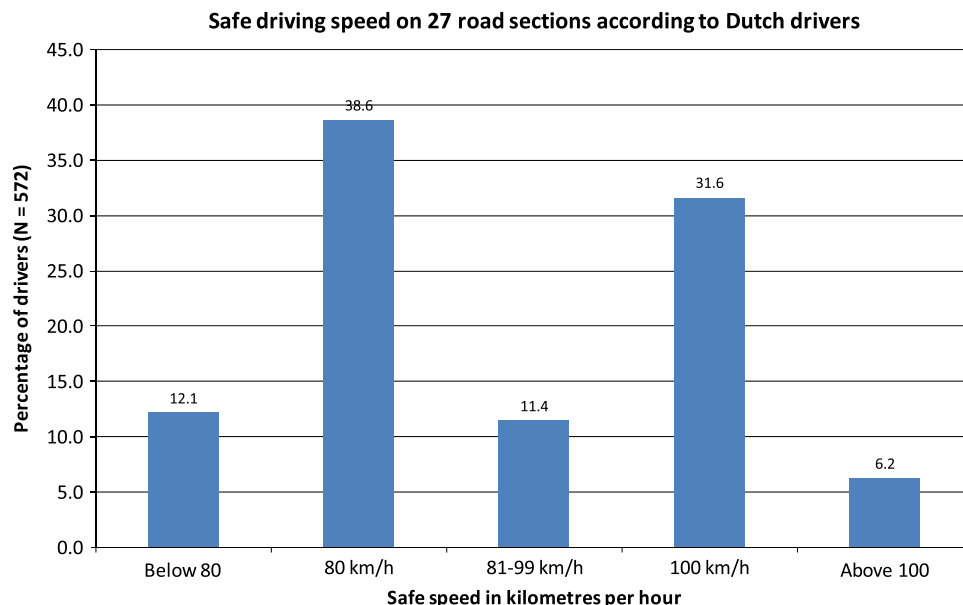


Fig. 4. Driver opinions regarding safe speed on a sample of 27 road sections that all had a speed limit of 80 km/h. Based on Golbenbeld and van Schagen (2007).

Table 4

Speed coordination as a game. Ordinal preferences (4=best; 1=worst). Outcomes for fast movers in upper right corner, outcomes for slow movers in lower left corner of each cell of the table.

		The fast movers	
		80 km/h	100 km/h
The slow movers	80 km/h	3 4	2 2
	100 km/h	1 1	4 3

other drivers to follow suit. Suddenly, everybody will be overtaking the slow driver and the speed of traffic will increase.

4. Implications of lack of rationality

Based on the research reviewed above, it seems clear that driver speed choice is not objectively rational. It is not even subjectively rational in all cases, since interaction with other drivers may prevent a driver from choosing his or her most preferred speed, forcing the driver to put up with driving at a speed that does not maximise his or her utility.

4.1. Main limits in rationality of driver speed choice

The lack of objective rationality in speed choice is attributable to the following cognitive limitations of driver speed choice:

1. Drivers tend to ignore, or assign minor importance to, impacts of speed that they do not immediately notice or that do not directly affect their personal utility. More specifically, environmental impacts of speed choice are largely ignored by drivers.
2. Drivers do not correctly perceive the relationship between speed and travel time. Gains in travel time attributable to small increases in high speed are overestimated, while corresponding gains attributable to small increases in low speed are underestimated. These misconceptions may lead drivers to commit more serious violations of low speed limits than of high speed limits.
3. Drivers underestimate the increase in the risk of accident associated with increased speed.
4. Drivers underestimate impact speed in situations in which it is clear that an accident is unavoidable, but its severity can be reduced by braking.
5. Driver preferences with regard to safe speed are very heterogeneous, making the coordination of speed choices difficult.

Ignoring environmental impacts of speed may impose significant external impacts in urban areas in terms of noise and air pollution. While these impacts cannot be entirely avoided, they can be minimised by setting speed limits at the speed that minimises the costs of noise and air pollution. In general, these speed limits are lower than the current mean speed of traffic in both urban and rural areas.

The fact that the risk of accidents, and injury severity, increases as speed increases has traditionally been an important

argument for introducing speed limits. It is, however, by itself not a convincing argument. If it is correct to assume that drivers choose the speed that is optimal, all things considered, the resulting number of accidents or injuries should also be regarded as optimal. This assumption is, however, not correct. Drivers tend to ignore some impacts of speed and they do not correctly estimate the impacts of speed choice on travel time or road safety. Misperception of the impacts of speed is reinforced by the fact that part of the cost of accidents is external from the driver's point of view (Elvik, 1994). The net result is that an "optimism bias" among drivers leads them to choose speeds that are higher than optimal from a societal point of view.

4.2. The calculus of misperceptions

The speed that minimises the sum of the costs of travel time (which, all else equal, are proportional to travel time) and accidents can be found by setting the sum of the first derivatives with respect to travel time and accident costs equal to zero. For the actual relationships, this comes to

$$\text{Optimal speed(actual relationships)} = -\frac{100}{v^2} + 3 \times 0.000001 \times v^2 = 0 \quad (5)$$

Optimal speed is estimated as 76 km/h. If, on the other hand, the perceived relationships, as modelled above, are used, optimal speed becomes

$$\text{Optimal speed(perceived relationships)} = [12.9 \times -0.0255 \times \exp(-0.0255 \times v)] + [1.2 \times 0.002 \times v^{0.2}] = 0 \quad (6)$$

Optimal speed is estimated to be above 130 km/h. This result lacks credibility and is clearly a product of the way the perceived relationships between speed and travel time and speed and relative accident cost have been parameterised. It can be argued that the misperception of the relationship between speed and accident risk is more severe than the misperception of the relationship between speed and travel time. The latter misperception can be corrected by providing drivers with feedback on mean speed and travel time when driving. Speed and travel time can be measured objectively. It is a lot more difficult to measure the level of accident risk in a direct way and give drivers credible feedback to inform them of the impacts of their choice of speed. The fact that the perceived relationship between speed and safety is considerably flatter than the actual relationship implies that the costs of increased speed will almost always be underestimated, in particular at high speeds.

Thus, the lack of rationality characterising driver speed choice would appear to be a strong argument for introducing speed limits. Any speed limits will, however, be a compromise between

multiple objectives that are partly conflicting. While safety is best served by low speed limits, reducing travel time is best served by high speed limits. On top of this, driver opinions with respect to speed limits are very heterogeneous. Thus, any speed limit is likely to be unpopular and regarded as either too high or too low by a considerable proportion of drivers.

4.3. The enforcement of speed limits

Even if drivers are less than perfectly rational when choosing speed, it does not necessarily follow that society would be better off by regulating speed choice by means of speed limits. For speed limits to be justified in terms of economic welfare theory, two conditions need to be fulfilled (Williamson, 2000):

1. Speed limits must influence actual driving speeds and bring them closer to optimal speeds, and
2. The costs of enforcing speed limits should not exceed the societal benefits of enforcement.

There are very few studies that have compared “free” driving speeds to those subject to speed limits. Most studies evaluating the effects of speed limits refer to changes from one speed limit to another (Elvik et al., 2004). These studies show that changing the speed limit will influence the mean speed of traffic. On the average, a 10 km/h change in speed limit will be associated with a 2.5 km/h change in mean speed. Thus, if, say, speed limit is lowered by 10 km/h, compliance the new speed limit will in general be lower than compliance with the old speed limit.

An interesting study was reported by Summala et al. (1980). They compared driving speeds before, during and after a 2-week police strike in Finland. During the strike, there was no enforcement of speed limits; speed was, in that sense, no longer effectively regulated by speed limits. The mean speed of traffic did increase by some 3 percent during the strike, but the rate of violations increased considerably more. Steinauer (1998) reports mean speeds close to 160 km/h on a section of a German Autobahn without a speed limit; this is a considerably higher mean speed than on Autobahn sections where an effectively enforced speed limit has been introduced (Lamm and Klöckner, 1984). It is therefore reasonable to believe that speed limits do influence speed and that the mean speed of traffic would be higher if speed limits did not exist.

The determination of optimal speed limits is, however, not an exact science. Elvik (2002) found that optimal speed limits are close to indeterminate for motorways and rural highways for speeds between 70 and 110 km/h. The variation in the total societal costs of travel in this range of speed is often less than 10 percent. A variation of this magnitude is definitely less than the uncertainty of the monetary valuation of the impacts of speeds. For motorways (freeways) in Norway, Elvik estimated the total costs of travel to NOK 2.91 at a speed of 80 km/h (1 NOK=0.17 US dollars as of December 2009), 2.82 at 90 km/h, 2.79 at 100 km/h, 2.80 at 110 km/h, and 2.85 at 120 km/h. The optimal speed limit is 100 km/h. But costs are only NOK 0.01 higher at 110 km/h. If a speed limit of 100 km/h needs to be enforced more vigorously than a speed limit of 110 km/h, this could tip the balance in favour of the latter speed limit. Thus, the costs of enforcement need to be considered when determining speed limits.

5. Conclusions

Speed is regulated by means of speed limits in all highly motorised countries. Speed limits would not be needed if drivers

were able to choose speeds that are optimal from a societal point of view without the guidance given by speed limits. Thus, the rationality of driver speed choice is an important criterion for assessing whether speed limits are justified or not in terms of normative welfare economics. Speed limits would not be justified if driver speed choice was perfectly rational and resulted in outcomes that are optimal from a societal point of view. This paper has reviewed research that has evaluated various aspects of the rationality of driver speed choice. A fairly strong case can be made that driver speed choice is not objectively rational and that the unaided coordination of speeds among drivers is difficult, because preferences regarding optimal speeds differ greatly. The lack of rationality in driver speed choice is a strong argument for the need for guiding and constraining this choice by means of speed limits.

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